Zihao Liu

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Education

University of Cambridge Cambridge, UK

BA (Hons) and MEng in Engineering (Year 3); ***Grade: II.1*** *Oct 2024 – Sept 2028*

Engineering Area: Information and Computer Engineering

Core modules:

- 3F8 **Inference**: regression, classification, PCA, hidden Markov models, gradient-based optimisation
- 3F3 **Statistical Signal Processing**: maximum-likelihood and Bayesian estimation, Wiener and matched filters
- 3M1 **Mathematical Methods**: SVD, least squares, Markov chains, constrained optimisation
- 3F7 **Information Theory and Coding**: entropy, source coding, error-correcting codes
- 3F1 **Signals and Systems**: z-transform, digital filtering, FFT, random processes
- 4M21 **Software Engineering and Design**: OOP, UML, design patterns, agile, HCI

Skills

Software Development: C/C++, **Java**, **Python**, **JavaScript**, TypeScript, Dart, Kotlin, Git, Linux, Angular, Flutter, Node.js, Spring Boot, AWS

Machine Learning & Data Science: NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, **PyTorch**, **TensorFlow**

Hardware & Engineering Tools: Arduino, SolidWorks, KiCad, LTSpice, PicoScope

Practices: Agile, Scrum, OOP, MVC and Design Patterns, DevOps, CI/CD

AI & LLM Systems:

- Built and shipped **agentic AI systems** on **agent harnesses**: RAG, **MCP** tool use, **Agent Skills**, and structured memory
- Engineered harness-level workflows via **context engineering**, **evals** (LLM-as-judge), and **guardrails**; accelerated development with **Claude Code** and **Codex**
- Research on **LLM inference accelerators** (in progress), and **LoRA-based federated learning** for **hallucination mitigation and defence**

Experience

Arm Cambridge, UK

Software Engineer, Central Engineering *Jun 2026 – Now, Full-time Internship*

- **Kernel, security, and isolation layer development** for Arm's next-generation CPU platform: low-level **C** firmware performing CPU integrity verification at power-on and establishing hardware-backed security layers prior to OS handoff.
- **Secure bootloader engineering**: cryptographically verify Linux image integrity and detect unauthorised modifications, safeguarding the trusted boot chain from power-on to Linux launch.
- **Low-level system and algorithm optimisation**: profiled and refined performance-critical code paths, substantially improving runtime efficiency across the pre-boot software stack.
- **Cross-team collaboration** within Arm's engineering workflows, leveraging **CI/CD** pipelines and standard testing tools to validate firmware reliability.

Minimax Labs Ltd. London, UK

Software Engineer *Jun 2025 – Sept 2025, Full-time Internship*

- Developed **frontend features** and improved **system reliability** for large-scale optimisation and data management platforms.
- Built reusable UI components in **TypeScript** and **Angular**, integrated with **Java** backends through **RESTful APIs**.
- Deepened software architecture knowledge through **OOP patterns**, dependency management, and UI/UX principles.

Cambridge AI Builder Club (Claude Builder Club) Cambridge, UK

President; Claude Campus Ambassador *Sept 2026 – Present*

- Lead the club and serve as **Claude Campus Ambassador at the University of Cambridge**.

Technical Officer *Oct 2025 – June 2026*

- **Workshop design and instruction**: prepared code labs, slides, and starter repositories; taught practical **LLM** skills including **prompt engineering**, MCP, Skills, **Claude API** integration, tool use/function calling, **RAG**, evaluation, and safety/guardrails.
- **Hackathon technical leadership**: produced scaffolds and baseline projects, set up API access and rate-limit strategies, mentored teams on-site, and coordinated judging criteria.
- **LLM demo development**: built demo apps and templates in **Python** and **React/Vue** with deep **Claude** integration.

Cambridge University Robotics Society Cambridge, UK

Software Engineer *Oct 2025 – June 2026*

- Developed control and perception systems for a **humanoid robot** based on **InMoov**.
- **Computer vision and LLM integration**: **hand gesture** and **pose detection** using Python, MediaPipe, and OpenCV, linked to LLM modules for natural interaction and command interpretation.
- **Embedded control** in **C++ and Arduino** for real-time gesture-based robot movement.

Cambridge University Chinese Undergraduate Society Cambridge, UK

DevOps Engineer; IT Department Coordinator *Oct 2024 – June 2026*

- Developed and maintained the Society's **website** (**React**) and **WeChat Mini Program** (**JavaScript**), improving engagement for **hundreds** of members.
- Implemented **DevOps** practices and managed the development workflow via **Git**, streamlining deployment for rapid, stable feature delivery.

Full Blue Racing Cambridge, UK

Electronics Engineer *Oct 2024 – June 2026*

- **Battery system design and specification** for a Formula Student electric race car: cell modules, BMS, and charging systems, enabling cell balancing, state-of-charge monitoring, and fault detection.
- Designed PCBs for the **MCU**, **ECU**, and **I²C serial bus** in KiCad, enabling real-time display of the car's operating status.
- **Embedded C/C++** for sensor data acquisition, I^2C communication, and system diagnostics on MCU and ECU.

Nanyang Technological University Singapore

Student Researcher *Apr 2022 – March 2023, Internship*

- Developed an **LSTM/GRU** framework for **Li-ion battery health prediction**, achieving an average RMSE of **0.724%** on real battery datasets (PyTorch); open-sourced on GitHub.
- Awarded **Gold** for *"Data-Driven Method for Li-Ion Battery Health Monitoring"*, supervised by Prof. Yan Xu, School of EEE, NTU.

Publications

Byzantine-Resilient Federated Large Language Model via Hypergraph Message Passing.
Kai Li, Hanlin Cai, **Zihao Liu**, Yichen Li, Jose Alcaraz Calero, Wei Ni, Falko Dressler, and Özgür B. Akan.
Under review at *IEEE Transactions on Information Forensics and Security*; arXiv:2610.12711, 2026.
Developed a hypergraph message-passing framework to detect and suppress malicious model updates in federated LLM training while preserving knowledge alignment across clients.

Awards & Achievements

Cambridge Trust Scholarship: Merit-based award supporting outstanding international students at the University of Cambridge. (Oct 2024 - Sept 2028)

Projects

Awesome Portfolio Page | *GitHub*

- Built a **React** portfolio template WebApp with data-driven configuration, deployable to GitHub Pages, Netlify, and Vercel.

Intelligent System to Prevent Contamination of Recyclables Jun 2023 – Sept 2023

- Created a YOLOv8-based smart bin on Raspberry Pi for accurate waste classification; **2nd runner-up**, *Sustain-a-City Challenge*.

DSTA Today-I-Learned (TIL) AI Hackathon | *GitHub* May 2022 – Jun 2022

- Trained CV and NLP models in PyTorch for robotic object detection and voice recognition; **2nd runner-up**, *DSTA TIL AI Competition*.